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Linear Discriminant Analysis (LDA)

Introduction to LDA

Linear Discriminant Analysis (LDA) is a supervised statistical method that fulfills two main objectives:

- **Dimensionality reduction** by finding the most discriminating directions
- **Classification** through probabilistic modeling of classes

Unlike Principal Component Analysis (PCA) which is unsupervised, LDA uses class labels to find directions that maximize separation between groups.

Supervised Learning Context

Formal Definition

In a supervised learning problem, we have:

- Data $X \subset \Omega \subset \mathbb{R}^D$
- Class labels $y = [M] \subset \mathbb{N}$

We seek to learn a function ϕ_θ that minimizes a loss $\mathcal{L}_{X \times y}(\theta)$ when predicting labels.

Examples of Supervised Methods

- Logistic regression
- Neural networks
- Support Vector Machines (SVM)

LDA distinguishes itself by its **generative** approach: it models the distribution of data within each class.

Geometric Intuition of LDA

The Fundamental Problem

Imagine points in two dimensions belonging to two classes.

How to find the **best line** on which to project these points to separate them as much as possible?

LDA answers this question by seeking a direction that:

1. **Separates class centers** (maximizes distance between means)
2. **Brings points closer** within each class (minimizes within-class variance)

Key Concepts: Between-Class and Within-Class Criteria

Basic Notations

For K classes:

- N_k : number of observations in class k
 - μ_k : mean of class k (D -dimensional vector)
 - μ : global mean
 - $x_i^{(k)}$: the i -th observation of class k
-

Between-Class Criterion

Intuition

The between-class criterion measures **how far apart the class centers are from each other**.

The more distant the class means are, the easier it will be to discriminate them.

Mathematical Formulation

Distance between two classes (binary case):

For two classes with means μ_1 and μ_2 , the squared distance between their projections on a direction a is:

$$\|\hat{\mu}_1 - \hat{\mu}_2\|^2 = \frac{(a^T(\mu_1 - \mu_2))^2}{\|a\|^2}$$

where $\hat{\mu}_k = \frac{a^T \mu_k}{\|a\|}$ is the projection of class k mean.

General case

The matrix captures the covariance of class centers relative to the global mean.

Normal Version (Summation)

Weighted Formulation

$$\Sigma_b = \sum_{k=1}^K N_k (\mu_k - \mu)(\mu_k - \mu)^T$$

Unweighted Formulation

$$\Sigma_b = \frac{1}{K} \sum_{k=1}^K (\mu_k - \mu)(\mu_k - \mu)^T$$

Matrix Version

Weighted Formulation

Let M be the $D \times K$ matrix where each column k is $\sqrt{N_k}(\mu_k - \mu)$:

$$M = [\sqrt{N_1}(\mu_1 - \mu) \quad \sqrt{N_2}(\mu_2 - \mu) \quad \cdots \quad \sqrt{N_K}(\mu_K - \mu)]$$

Then:

$$\Sigma_b = MM^T$$

Unweighted Formulation

Let M' be the $D \times K$ matrix where each column k is $(\mu_k - \mu)$:

$$M' = [(\mu_1 - \mu) \quad (\mu_2 - \mu) \quad \cdots \quad (\mu_K - \mu)]$$

Then:

$$\Sigma_b = \frac{1}{K} M' M'^T$$

Equivalence between Formulations

Relation

The two formulations give the same matrix up to a scaling factor:

$$MM^T = \left(\sum_{k=1}^K N_k \right) \cdot \frac{1}{K} M' M'^T$$

when $N_1 = N_2 = \dots = N_K$ (balanced classes).

For LDA

This scaling difference does not affect the discriminant directions, only their amplitude in Fisher's criterion.

Geometric Interpretation

Σ_b is a $D \times D$ matrix that measures:

- How class centers vary around the global mean
- The dispersion “between” different classes

The quantity $a^T \Sigma_b a$ represents the **projected between-class variance** on direction a .

Important Property

For K classes, the matrix Σ_b has rank at most $K - 1$.

This means there are at most $K - 1$ independent directions that separate the classes.

Within-Class Criterion

Intuition

The within-class criterion measures **how closely points within the same class are grouped**. The more concentrated the points of a class are, the more homogeneous the class is and the easier to identify.

Mathematical Formulation

For each class k , we define its covariance matrix:

$$\Sigma_k = \frac{1}{N_k} \sum_{i=1}^{N_k} (x_i^{(k)} - \mu_k)(x_i^{(k)} - \mu_k)^T$$

Total within-class dispersion matrix:

$$\Sigma_w = \sum_{k=1}^K \Sigma_k = \sum_{k=1}^K \sum_{i=1}^{N_k} (x_i^{(k)} - \mu_k)(x_i^{(k)} - \mu_k)^T$$

Interpretation

Σ_w measures the **overall dispersion within all classes**.

A small value of $a^T \Sigma_w a$ indicates that the projections of points are clustered around their class mean.

Fisher's Criterion: Combination of Both Criteria

Principle

We seek a direction a that:

- Maximizes between-class dispersion: $a^T \Sigma_b a$ (large)
- Minimizes within-class dispersion: $a^T \Sigma_w a$ (small)

Fisher's criterion combines these two objectives into a single ratio:

$$J(a) = \frac{a^T \Sigma_b a}{a^T \Sigma_w a}$$

Ratio Interpretation

$J(a)$ represents the **signal-to-noise ratio** for discrimination:

- **Numerator:** “signal” = separation between classes
- **Denominator:** “noise” = variability within classes

The larger $J(a)$, the better direction a is for discriminating classes.

Optimization

We seek the direction a^* that maximizes $J(a)$:

$$a^* = \arg \max_a \frac{a^T \Sigma_b a}{a^T \Sigma_w a}$$

This optimization problem has an analytical solution.

Projection of Between and Within-Class Criteria

When projecting onto a direction a (not necessarily unit), the projected statistics include a normalization factor.

Projection of Centers

For a direction a , the projection of a vector v is given by:

$$\hat{v} = \frac{a^T v}{\|a\|}$$

The projected class means become:

$$\hat{\mu}_k = \frac{a^T \mu_k}{\|a\|}, \quad \hat{\mu} = \frac{a^T \mu}{\|a\|}$$

Projected Between-Class Criterion

The projected between-class variance is expressed as:

$$\frac{1}{M} \sum_{k=1}^M (\hat{\mu}_k - \hat{\mu})^2 = \frac{1}{\|a\|^2} a^T \left[\frac{1}{M} \sum_{k=1}^M (\mu_k - \mu)(\mu_k - \mu)^T \right] a$$

Simply:

$$\frac{1}{M} \sum_{k=1}^M (\hat{\mu}_k - \hat{\mu})^2 = \frac{1}{\|a\|^2} a^T \Sigma_b a$$

Projected Within-Class Criterion

For a class k , the projected variance is:

$$\hat{\sigma}_k^2 = \frac{1}{N_k} \sum_{i=1}^{N_k} (\hat{x}_i^{(k)} - \hat{\mu}_k)^2 = \frac{1}{\|a\|^2} a^T \Sigma_k a$$

The total projected within-class variance becomes:

$$\sum_{k=1}^M \hat{\sigma}_k^2 = \frac{1}{\|a\|^2} a^T \left(\sum_{k=1}^M \Sigma_k \right) a = \frac{1}{\|a\|^2} a^T \Sigma_w a$$

Consequence for Fisher's Criterion

The projected Fisher ratio is written:

$$J(a) = \frac{\text{Projected between-class variance}}{\text{Projected within-class variance}} = \frac{\frac{1}{\|a\|^2} a^T \Sigma_b a}{\frac{1}{\|a\|^2} a^T \Sigma_w a}$$

The factors $\frac{1}{\|a\|^2}$ cancel out:

$$J(a) = \frac{a^T \Sigma_b a}{a^T \Sigma_w a}$$

Conclusion: Fisher's criterion depends only on the **direction** of a , not its norm.
We can therefore search for a without normalization constraint, then normalize if necessary.

Optimization Problem Solution

Mathematical Derivation

To maximize $J(a)$, we set its gradient to zero:

$$\frac{\partial J(a)}{\partial a} = \frac{2\Sigma_b a(a^T \Sigma_w a) - 2\Sigma_w a(a^T \Sigma_b a)}{(a^T \Sigma_w a)^2} = 0$$

Which implies:

$$\Sigma_b a = \frac{a^T \Sigma_b a}{a^T \Sigma_w a} \Sigma_w a$$

Noting $\lambda = \frac{a^T \Sigma_b a}{a^T \Sigma_w a} = J(a)$, we obtain:

$$\Sigma_b a = \lambda \Sigma_w a$$

This is a **generalized eigenvalue problem**.

Eigenvalue Solution

The solution is given by the eigenvectors of $\Sigma_w^{-1}\Sigma_b$:

$$\Sigma_w^{-1}\Sigma_b a = \lambda a$$

Important properties:

- There are at most $K - 1$ non-zero eigenvalues
- The eigenvectors corresponding to the largest eigenvalues are the most discriminating directions
- The largest eigenvalue λ_1 gives the maximum of $J(a)$

Special Case: Two Classes

For $K = 2$, there is only one discriminating direction:

$$a \propto \Sigma_w^{-1}(\mu_1 - \mu_2)$$

This direction is proportional to the difference of means, “corrected” by the inverse covariance.

Generalization to Multiple Dimensions

Discriminant Subspaces

For K classes, we can obtain up to $K - 1$ discriminating directions:

- a_1 : most discriminating direction (largest eigenvalue λ_1)
- a_2 : second most discriminating direction ($\lambda_2 < \lambda_1$)
- ...
- a_{K-1} : last discriminating direction

Projection onto Discriminant Subspace

The projection matrix $W = [a_1, a_2, \dots, a_{K-1}]$ allows reducing data dimension from D to $K - 1$:

$$Y = XW$$

where Y are the data projected into the new space.

Eigenvalue Interpretation

The eigenvalue λ_i measures the **discriminant power** of direction a_i :

- Large λ_i : strong separation between classes along a_i
- Small λ_i : weak separation

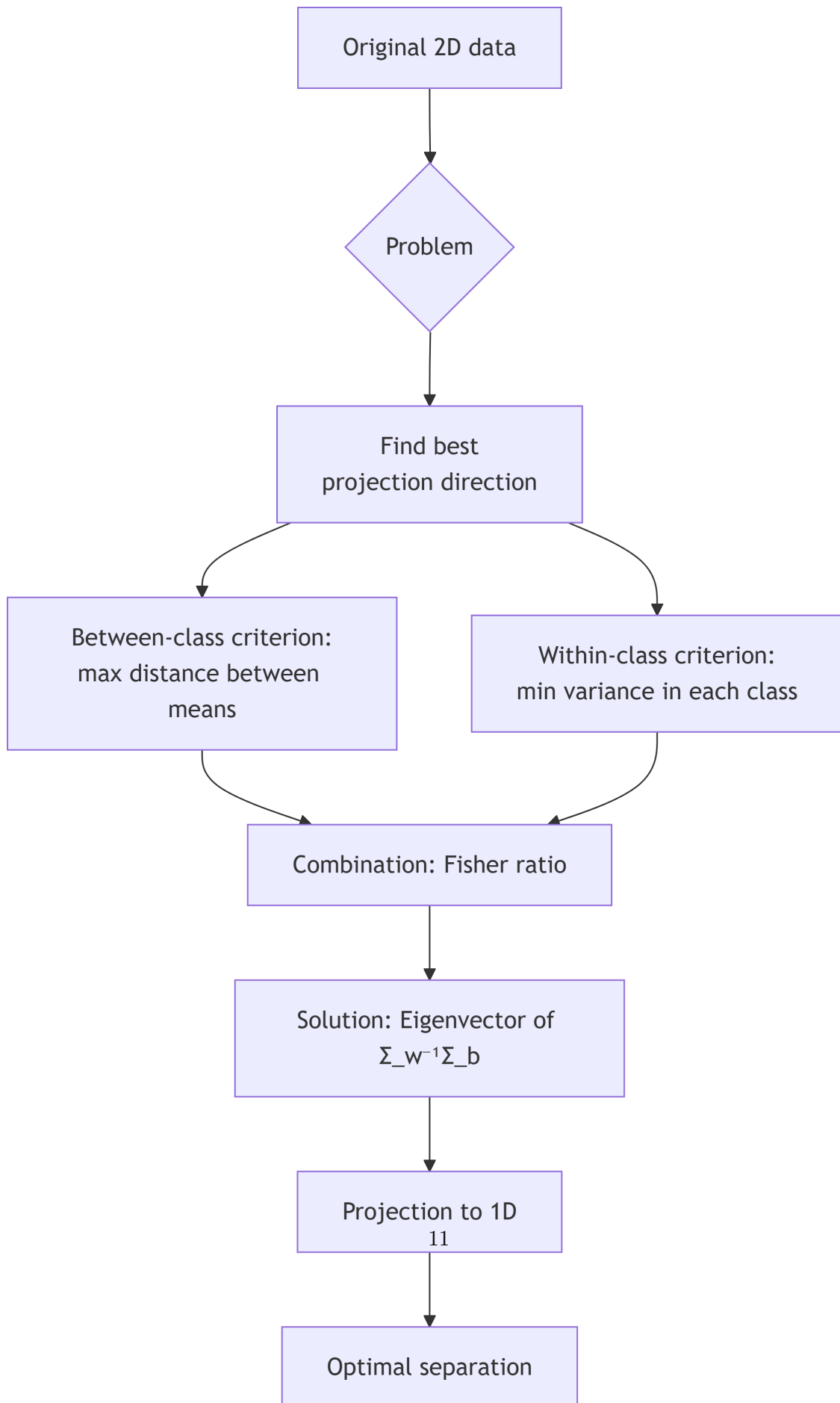
The percentage of total discrimination explained by direction a_i is:

$$\text{PVE}_i = \frac{\lambda_i}{\sum_{j=1}^{K-1} \lambda_j} \times 100\%$$

Visualization of Concepts

Two-Dimensional Example

Consider two classes with Gaussian distributions:



Geometric Interpretation

The optimal direction a is the one that, when data is projected onto it, produces:

- Well-separated histograms for each class
 - Distributions with little overlap
 - Maximum distance between means relative to internal variability
-

Conditional modeling in supervised classification

In **supervised classification** problems, the objective is to predict the class label y associated with an observation x .

A fundamental approach consists of modeling the **conditional probability** $P(y | x)$, i.e., the probability that an observation x belongs to a given class y .

Bayesian framework for classification

Bayes rule and optimal decision

Bayes' theorem allows expressing the **posterior probability** that an observation x belongs to class k as follows:

$$P(C = k | x) = \frac{P(x | C = k) \cdot P(C = k)}{P(x)}$$

Notations:

- $P(C = k)$: **Prior probability** of class k .
- $P(x | C = k)$: **Likelihood** of x given class k .
- $P(x)$: **Marginal probability** of x (constant for all classes).

The **optimal decision rule** (minimizing error rate) consists of assigning x to the class k maximizing the posterior probability:

$$\delta(x) = \arg \max_k P(C = k | x)$$

Since $P(x)$ is constant, this rule simplifies to:

$$\delta(x) = \arg \max_k [P(x | C = k) \cdot P(C = k)]$$

Linear Discriminant Analysis (LDA) as a conditional generative model

Fundamental assumptions of LDA

Linear Discriminant Analysis (LDA) relies on three key assumptions:

1. **Gaussian distribution per class:**

Each class follows a multivariate normal law:

$$P(x | C = k) \sim \mathcal{N}(\mu_k, \Sigma)$$

where μ_k is the mean vector of class k , and Σ the covariance matrix.

2. **Common covariance matrix:**

All classes share the **same covariance matrix** Σ .

3. **Prior probabilities:**

Classes are characterized by prior probabilities $P(C = k)$, often estimated by the observed proportions in the sample.

Explicit formulation of the LDA model

Under these assumptions, the probability density for class k is written:

$$P(x | C = k) = \frac{1}{(2\pi)^{D/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu_k)^T \Sigma^{-1} (x - \mu_k) \right)$$

where:

- D : Data dimension.
 - μ_k : Mean vector of class k .
 - Σ : Common covariance matrix.
-

Derivation of the linear discriminant function

Transition to decision function

Applying the Bayesian decision rule and taking the **logarithm** (which preserves maxima), we obtain:

$$\delta(x) = \arg \max_k [\ln P(x | C = k) + \ln P(C = k)]$$

By developing the quadratic term and eliminating constants independent of k , the discriminant function simplifies to:

$$\delta(x) = \arg \max_k \left[x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \ln P(C = k) \right]$$

Linear form of the decision boundary

The discriminant function for class k can be expressed as a **linear function** in x :

$$g_k(x) = w_k^T x + b_k$$

where:

- $w_k = \Sigma^{-1} \mu_k$: Weight vector for class k .
- $b_k = -\frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \ln P(C = k)$: Bias term.

Linear decision boundary:

The boundary between two classes k and l is defined by the equation:

$$\{x \mid g_k(x) = g_l(x)\} \quad \Rightarrow \quad (w_k - w_l)^T x + (b_k - b_l) = 0$$

Advantages and Limitations

Advantages of LDA

- **Solid theoretical foundations** based on Bayesian statistics
- **Analytical solution** (no iterative optimization)
- **Efficient** when assumptions are respected
- **Interpretable**: discriminant directions have clear meaning
- **Dual use**: dimensionality reduction and classification

Limitations

- **Normality assumption:** requires approximately Gaussian data
- **Homoscedasticity:** assumes same covariance for all classes
- **Sensitivity to outliers**
- **Linearity:** does not capture non-linear relationships
- **Problems in high dimension** when $D > N$

Alternatives and Extensions

QDA (Quadratic Discriminant Analysis)

Relaxes the common covariance assumption, allowing quadratic decision boundaries.

Regularized LDA

Adds regularization to stabilize the estimation of Σ^{-1} in high dimension.

Kernel LDA

Uses kernel functions to capture non-linear relationships.

Flexible Discriminant Analysis

Models conditional distributions using non-parametric methods.

Comparison with Other Methods

LDA vs PCA

Aspect	LDA	PCA
Supervision	Supervised (uses labels)	Unsupervised
Objective	Maximize separation between classes	Maximize total variance
Directions	Discrimination directions	Maximum variance directions
Interpretation	Separation of groups	Variability structure

LDA vs Logistic Regression

Aspect	LDA	Logistic Regression
Approach	Generative model	Discriminative model
Assumptions	Normal with common covariance	None on distribution of x
Stability	More stable with few data	May require more data
Boundaries	Always linear under its assumptions	Linear by construction

Practical Applications

Application Domains

Biology and Medicine

- Cancer cell classification
- Diagnosis from medical images
- Genomic data analysis

Finance

- Credit scoring
- Fraud detection
- Customer classification

Computer Vision

- Face recognition
- Image classification
- Handwriting recognition

Marketing

- Customer segmentation
- Preference analysis
- Product recommendation

Best Practices

Validation of assumptions

- Test data normality (Shapiro-Wilk test, QQ-plots)
- Check covariance homogeneity (Box's test)
- Examine outliers

Data preprocessing

- Variable standardization
- Missing value management
- Class balancing if necessary

Model validation

- Cross-validation
 - Adapted performance measures (confusion matrix, AUC, etc.)
 - Comparison with reference methods
-

Conclusion

Linear Discriminant Analysis is an **elegant and powerful** method based on solid statistical principles.

Its essence lies in the joint optimization of two criteria:

Between-class criterion

Maximize separation between class centers, making classes as distinct as possible.

Within-class criterion

Minimize dispersion within each class, making each class as coherent as possible.

The combination of these two objectives via **Fisher's criterion** allows finding optimal directions for discriminating classes while reducing data dimensionality.

Although based on sometimes restrictive assumptions, LDA remains a valuable tool in the data scientist's toolbox, particularly useful for:

- Exploration and visualization of labeled data
- Classification when assumptions are reasonably verified
- Understanding discriminating variables in a problem

Its interpretability and conceptual simplicity make it an excellent introduction to supervised classification and dimensionality reduction methods.